

Lesson 28: Plane Curves and Parametric Equations

Criteria	Proficient (High)	Developing (Mid)	Beginning (Low)
Sketching Curves	Correctly plots points using t values, connecting them in a smooth curve.	Plots most points correctly but may have minor errors in shape or orientation.	Struggles to calculate x(t) and y(t) points or plots points incorrectly.
Eliminating Parameter	Successfully eliminates parameter to form a correct Cartesian equation $f(x) = 0$.	Eliminates the parameter but makes algebraic errors in simplification.	Cannot manipulate equations to isolate and substitute t.
Orientation & Direction	Consistently marks arrows indicating the path of the curve as t increases.	Marks orientation but omits or incorrectly identifies direction.	Fails to indicate direction of motion on the graph.
Domain & Range	Correctly identifies the domain and range of the parametric equations.	Identifies parts of the domain/range but misses constraints based on t,	Does not identify constraints on x or y.